ABACUS / IMPLEMENTATION 1.0

Final verification report

Executed checks, 15-phase traceability and outstanding MATLAB acceptance

16 SEPTEMBER 2026

Implementation and recorded host checks. MATLAB, GUI, Simulink and ML training have not executed on the delivery host.

FINAL VERIFICATION REPORT

Project: Aircraft Engine Performance Analysis, Digital Twin, Machine Learning and Intelligent Engine Selection System (ABACUS v1.1 implementation).

Delivery date: 2026-09-16 UTC.

Decision

WARNING - implementation delivered; full MATLAB runtime acceptance remains outstanding.

All 15 requested implementation areas are represented by integrated source, relevant tests and documentation. No phase is called complete solely because files exist. The host actually executed 28 SQL/static checks: 28 PASS, 0 FAIL. There are 93 MATLAB source files and 64 test functions. MATLAB/Octave, the GUI, Simulink and ML training were unavailable and are NOT EXECUTED. No measured performance, trained model, accuracy or validation success is fabricated.

The supplied database is a real SQLite file with reproducible schema/seed, not an empty sample. Both production and test copies passed integrity and foreign-key checks. The default seed contains 11 engines, 192 matching numeric provenance rows, 13 unverified secondary reference points and 7 missions. Optional piston seeding was tested through host SQLite and yields 13 engines.

Status definitions

- PASS: the named, narrowly scoped check actually executed and passed.
- WARNING: implementation or evidence exists, but a material limitation or acceptance requirement remains.
- NOT EXECUTED: the stated runtime/test was not available or run.

Executed checks

Check: Fresh schema and seed execute	Status: PASS
Check: Eleven core engines	Status: PASS
Check: Required tables, metadata, views and indexes	Status: PASS
Check: Reproducible idempotent seed	Status: PASS
Check: Foreign keys explicitly enabled	Status: PASS
Check: Foreign-key violations rejected	Status: PASS
Check: Database integrity and FK checks	Status: PASS
Check: Every numeric parameter has matching provenance and unit	Status: PASS
Check: No unverified manufacturer/type-certificate claims	Status: PASS
Check: Kaveri provenance remains Low/Estimated or Assumed	Status: PASS
Check: Missing TSFC not fabricated	Status: PASS
Check: All four data types accepted, invalid enum rejected	Status: PASS

Check: Reference records immutable | Status: PASS

Check: Simulation log append-only | Status: PASS

Check: Transaction rollback | Status: PASS

Check: Quoted text survives prepared-value roundtrip | Status: PASS

Check: Efficiency, OPR and mission weight constraints | Status: PASS

Check: Six PDF missions plus requested Trainer | Status: PASS

Check: Optional piston seed is separate and reproducible | Status: PASS

Check: Delivered engine_analysis.db integrity | Status: PASS

Check: Delivered engine_test.db integrity | Status: PASS

Check: Unique public MATLAB filenames | Status: PASS

Check: No forbidden unfinished implementation markers | Status: PASS

Check: No hardcoded author workspace paths in MATLAB | Status: PASS

Check: MATLAB source entry names match filenames | Status: PASS

Check: GUI dispatched action registry has handlers | Status: PASS

Check: Simulink wrapper and model references exist | Status: PASS

Check: Regression and ML gates are implemented | Status: PASS

The audit evidence, timestamp and exact scope are in reports/HOST_AUDIT_RESULTS.json. Source parsing evidence is in reports/SOURCE_PARSE_RESULTS.json. A syntax/static pass does not prove numerical correctness or MATLAB API execution.

Included test inventory

Suite: TestAtmosphere.m | Tests: 8 | Execution: NOT EXECUTED

Suite: TestDatabase.m | Tests: 10 | Execution: NOT EXECUTED

Suite: TestEngines.m | Tests: 16 | Execution: NOT EXECUTED

Suite: TestGUI.m | Tests: 1 | Execution: NOT EXECUTED

Suite: TestIntegration.m | Tests: 12 | Execution: NOT EXECUTED

Suite: TestML.m | Tests: 8 | Execution: NOT EXECUTED

Suite: TestMission.m | Tests: 8 | Execution: NOT EXECUTED

Suite: TestSimulink.m | Tests: 1 | Execution: NOT EXECUTED

The full test names appear in docs/TESTING.md. Tests cover atmosphere layers/bounds, SQL integrity/rollback/provenance, energy/mass balance, nozzle choking, engine zero-throttle and invalid inputs, grid shape, comparison compatibility, mission hard filtering and tie cases, ML splitting/metrics/domain/gates, GUI callbacks and Simulink-backend agreement. Some algorithm API tests use simple synthetic mathematical arrays; those are test fixtures, not engine performance claims.

Delivery limitations

The GUI is a programmatic AppBase .m application, not a .mlapp Design View archive. Simulink includes three .mdl definitions plus an executable .slx builder; no .slx has been generated here. All engine seed parameters remain Estimated/Assumed and Low; none was independently manufacturer-verified. Default ML training is blocked until actual physics tests and

source-reviewed, in-tolerance Track A comparisons pass. No pretrained models or fabricated certificates are included.

Steady-state constant-property models do not include afterburners, detailed component/propeller maps, bleed/cooling, variable geometry, degradation, transients, hardware, certification or aircraft drag/range. The technical report documents implementation and source, not a completed experimental validation study. Source and GitHub-ready metadata are delivered without an invented repository history.

Next acceptance commands

Run from the extracted project root in MATLAB R2023b or newer:

```
setup_project;
verify_project;
run_all_tests;
validate_physics;
run_project('gui');
```

Review actual results and unresolved reference errors before ML training. Install Simulink before build_simulink_models and run_simulink_model. See README.md for the complete workflow.

Per-phase traceability

Statuses distinguish implementation from execution. No entire physics, ML or GUI phase is marked PASS based on source presence.

Phase: 1 | Area: Database and project infrastructure | Status: PASS for executed host SQL subchecks; WARNING for phase acceptance
Phase: 2 | Area: ISA atmosphere | Status: NOT EXECUTED
Phase: 3 | Area: Piston physics | Status: NOT EXECUTED
Phase: 4 | Area: Turbojet physics | Status: NOT EXECUTED
Phase: 5 | Area: Turbofan physics | Status: NOT EXECUTED
Phase: 6 | Area: Turboprop physics | Status: NOT EXECUTED
Phase: 7 | Area: Physics validation and automated testing | Status: WARNING
Phase: 8 | Area: Digital Twin coordinator | Status: NOT EXECUTED
Phase: 9 | Area: Performance grids | Status: NOT EXECUTED
Phase: 10 | Area: Comparison and analysis | Status: NOT EXECUTED
Phase: 11 | Area: Mission-based selection | Status: NOT EXECUTED
Phase: 12 | Area: ML dataset generation | Status: NOT EXECUTED
Phase: 13 | Area: ML surrogate training and evaluation | Status: NOT EXECUTED
Phase: 14 | Area: MATLAB GUI | Status: NOT EXECUTED
Phase: 15 | Area: Full integration and reports | Status: WARNING

PHASE 1 - Database and project infrastructure

Status: PASS for executed host SQL subchecks; WARNING for phase acceptance

Files: setup_project.m, project_paths.m, matlab/setup, database/schema.sql, database/seed_data.sql, database/engine_analysis.db, database/engine_test.db.

Dependencies: MATLAB R2023b, Database Toolbox.

Included test suites: TestDatabase.m.

Actually executed / reviewed: 28 host audit checks include real SQLite creation, seed, constraints, rollback, provenance and delivered-file integrity. MATLAB native connection/write APIs are NOT EXECUTED.

Known limitations and unresolved issues: Native SQLite typed operations replace JDBC-specific examples. Runtime API acceptance remains open.

Test cases in the applicable shared suites:

- TestDatabase.m: testAllTablesAndEngines, testSeedIdempotent, testProvenanceAudit, testForeignKeyFailure, testLiteralTextRoundTrip, testAtomicRollback, testImmutableReference, testMissingProvenanceRejected, testInvalidEngineID, testSchemaConstraints.

PHASE 2 - ISA atmosphere

Status: NOT EXECUTED

Files: matlab/atmosphere/ISA_model.m, matlab/utilities/flight_condition.m.

Dependencies: project_config.

Included test suites: TestAtmosphere.m.

Actually executed / reviewed: Static source review only. No MATLAB atmosphere results were executed.

Known limitations and unresolved issues: Geometric/geopotential convention and inconsistent blueprint checkpoints are explicitly corrected.

Test cases in the applicable shared suites:

- TestAtmosphere.m: testSeaLevel, testGeopotentialCheckpoints, testLayerContinuity, testGeometricConversion, testMachStagnation, testNegativeAltitude, testUpperBound, testInvalidMachAndThrottle.

PHASE 3 - Piston physics

Status: NOT EXECUTED

Files: matlab/engines/piston/Piston_Model.m, database/seed_piston_extension.sql, matlab/setup/db_seed_piston_extension.m.

Dependencies: Phase 1, Phase 2, propeller_model.

Included test suites: TestEngines.m.

Actually executed / reviewed: Host SQL check confirmed optional seed yields 13 engines. MATLAB physics was not executed.

Known limitations and unresolved issues: Opt-in extension outside PDF core; assumed naturally aspirated parameters.

Test cases in the applicable shared suites:

- TestEngines.m: testIdealCompressionCheckpoint, testNozzleChoked, testNozzleUnchoked, testNozzleBoundaryContinuity, testNoExpansionRejected, testAllCoreEnginesSLS, testTurbofanMassAndFanDemand, testTurbopropPowerAndStaticCase, testPropellerContinuousAndFinite, testKaveriWarningAndType, testAltitudeAndMachEnvelope, testAfterburnerRejected, testBadEfficiencyAndTemperature, testPistonPowerTorqueAndAltitude, testZeroThrottleState, testChangedParametersInvalidateFingerprint.

PHASE 4 - Turbojet physics

Status: NOT EXECUTED

Files: matlab/engines/turbojet/TJ_Model.m, matlab/engines/common.

Dependencies: Phase 1, Phase 2.

Included test suites: TestEngines.m.

Actually executed / reviewed: Common cycle/component source reviewed; MATLAB mass/energy and nozzle tests unexecuted.

Known limitations and unresolved issues: Constant properties and algebraic off-design flow; no afterburner.

Test cases in the applicable shared suites:

- TestEngines.m: testIdealCompressionCheckpoint, testNozzleChoked, testNozzleUnchoked, testNozzleBoundaryContinuity, testNoExpansionRejected, testAllCoreEnginesSLS, testTurbofanMassAndFanDemand, testTurbopropPowerAndStaticCase, testPropellerContinuousAndFinite, testKaveriWarningAndType, testAltitudeAndMachEnvelope, testAfterburnerRejected, testBadEfficiencyAndTemperature, testPistonPowerTorqueAndAltitude, testZeroThrottleState, testChangedParametersInvalidateFingerprint.

PHASE 5 - Turbofan physics

Status: NOT EXECUTED

Files: matlab/engines/turbofan/TF_Model.m, matlab/engines/common/gas_turbine_cycle.m.

Dependencies: Phase 4 components.

Included test suites: TestEngines.m.

Actually executed / reviewed: Static station, split-flow and work-demand review; no MATLAB execution.

Known limitations and unresolved issues: Separate exhaust, corrected fan work, no mixer or compressor map.

Test cases in the applicable shared suites:

- TestEngines.m: testIdealCompressionCheckpoint, testNozzleChoked, testNozzleUnchoked, testNozzleBoundaryContinuity, testNoExpansionRejected, testAllCoreEnginesSLS, testTurbofanMassAndFanDemand, testTurbopropPowerAndStaticCase, testPropellerContinuousAndFinite, testKaveriWarningAndType, testAltitudeAndMachEnvelope, testAfterburnerRejected, testBadEfficiencyAndTemperature, testPistonPowerTorqueAndAltitude, testZeroThrottleState, testChangedParametersInvalidateFingerprint.

PHASE 6 - Turboprop physics

Status: NOT EXECUTED

Files: matlab/engines/turboprop/TP_Model.m, matlab/engines/common/propeller_model.m.

Dependencies: Phase 4 components.

Included test suites: TestEngines.m.

Actually executed / reviewed: Static gas-generator/free-turbine/propeller review; no MATLAB power/reference result.

Known limitations and unresolved issues: Assumed residual pressure and propeller geometry. Rating agreement unresolved.

Test cases in the applicable shared suites:

- TestEngines.m: testIdealCompressionCheckpoint, testNozzleChoked, testNozzleUnchoked, testNozzleBoundaryContinuity, testNoExpansionRejected, testAllCoreEnginesSLS, testTurbofanMassAndFanDemand, testTurbopropPowerAndStaticCase, testPropellerContinuousAndFinite, testKaveriWarningAndType, testAltitudeAndMachEnvelope, testAfterburnerRejected, testBadEfficiencyAndTemperature, testPistonPowerTorqueAndAltitude, testZeroThrottleState, testChangedParametersInvalidateFingerprint.

PHASE 7 - Physics validation and automated testing

Status: WARNING

Files: matlab/validation, run_all_tests.m, tests.

Dependencies: Phases 1-6, matlab.unittest.

Included test suites: TestAtmosphere.m, TestEngines.m, TestDatabase.m.

Actually executed / reviewed: Host audit executed. MATLAB validation, sensitivity and regression baseline have NOT EXECUTED.

Known limitations and unresolved issues: No supplied verified primary engine targets, no accepted validation certificates.

Test cases in the applicable shared suites:

- TestAtmosphere.m: testSeaLevel, testGeopotentialCheckpoints, testLayerContinuity, testGeometricConversion, testMachStagnation, testNegativeAltitude, testUpperBound, testInvalidMachAndThrottle.
- TestEngines.m: testIdealCompressionCheckpoint, testNozzleChoked, testNozzleUnchoked,

testNozzleBoundaryContinuity, testNoExpansionRejected, testAllCoreEnginesSLS, testTurbofanMassAndFanDemand, testTurbopropPowerAndStaticCase, testPropellerContinuousAndFinite, testKaveriWarningAndType, testAltitudeAndMachEnvelope, testAfterburnerRejected, testBadEfficiencyAndTemperature, testPistonPowerTorqueAndAltitude, testZeroThrottleState, testChangedParametersInvalidateFingerprint.

- TestDatabase.m: testAllTablesAndEngines, testSeedIdempotent, testProvenanceAudit, testForeignKeyFailure, testLiteralTextRoundTrip, testAtomicRollback, testImmutableReference, testMissingProvenanceRejected, testInvalidEngineID, testSchemaConstraints.

PHASE 8 - Digital Twin coordinator

Status: NOT EXECUTED

Files: matlab/digital_twin/DT_Coordinator.m, matlab/digital_twin/simulate_engine.m, matlab/setup/db_store_results.m.

Dependencies: Phases 1-7, ML predictor for L5.

Included test suites: TestIntegration.m.

Actually executed / reviewed: Static L1-L5 dispatch/persistence/fallback review. MATLAB integration tests unexecuted.

Known limitations and unresolved issues: Academic steady-state coordinator; no live sensor connection.

Test cases in the applicable shared suites:

- TestIntegration.m: testPersistenceAndLog, testFailedPointLogsWithoutPerformance, testLogImmutable, testArchitectureGridCounts, testGridValidityMaskAndTemperatureFields, testCompatibleComparison, testIncompatibleComparisonRejected, testAllMissionPresets, testUnverifiedReferencesCannotPass, testReportingContainsProvenance, testOutputUnitConversion, testHTMLTextEscaped.

PHASE 9 - Performance grids

Status: NOT EXECUTED

Files: matlab/performance/performance_grid.m, matlab/performance/evaluate_grid.m.

Dependencies: Phase 8.

Included test suites: TestIntegration.m.

Actually executed / reviewed: Configured cardinalities/source checked; full engine grids not generated.

Known limitations and unresolved issues: Invalid cells retained and excluded from training; some grid knots exceed an engine envelope by design.

Test cases in the applicable shared suites:

- TestIntegration.m: testPersistenceAndLog, testFailedPointLogsWithoutPerformance, testLogImmutable, testArchitectureGridCounts, testGridValidityMaskAndTemperatureFields,

testCompatibleComparison, testIncompatibleComparisonRejected, testAllMissionPresets, testUnverifiedReferencesCannotPass, testReportingContainsProvenance, testOutputUnitConversion, testHTMLTextEscaped.

PHASE 10 - Comparison and analysis

Status: NOT EXECUTED

Files: matlab/comparison, matlab/reporting/plot_performance.m, matlab/reporting/plot_performance_map.m.

Dependencies: Phase 8, Phase 9.

Included test suites: TestIntegration.m.

Actually executed / reviewed: Static metric compatibility and sorting review; numerical tables/plots not executed.

Known limitations and unresolved issues: No unconditional comparison of jet TSFC to propeller BSFC.

Test cases in the applicable shared suites:

- TestIntegration.m: testPersistenceAndLog, testFailedPointLogsWithoutPerformance, testLogImmutable, testArchitectureGridCounts, testGridValidityMaskAndTemperatureFields, testCompatibleComparison, testIncompatibleComparisonRejected, testAllMissionPresets, testUnverifiedReferencesCannotPass, testReportingContainsProvenance, testOutputUnitConversion, testHTMLTextEscaped.

PHASE 11 - Mission-based selection

Status: NOT EXECUTED

Files: matlab/mission.

Dependencies: Phases 1,8,10.

Included test suites: TestMission.m.

Actually executed / reviewed: Host audit verifies seven mission seeds; MATLAB constraint/scoring tests unexecuted.

Known limitations and unresolved issues: Relative engine screening only; no airframe drag/range solver.

Test cases in the applicable shared suites:

- TestMission.m: testNormalisationDirections, testTieAndSingleMetric, testAllEliminated, testOnlyOneSurvives, testHighScoreCannotOverrideConstraint, testArchitectureAndCeiling, testWeightsRejected, testMixedSFCRejected.

PHASE 12 - ML dataset generation

Status: NOT EXECUTED

Files: matlab/ml/ml_generate_dataset.m, matlab/ml/ml_stratified_split.m, matlab/ml/ml_targets.m.

Dependencies: Phase 7 acceptance, Phase 9.

Included test suites: TestML.m.

Actually executed / reviewed: Static gate/split/domain review. No accepted physics dataset generated.

Known limitations and unresolved issues: Default validation gate closed. Synthetic physics data only.

Test cases in the applicable shared suites:

- TestML.m: testSplitDeterministicAndDisjoint, testDuplicateInputsRejected, testMetricsKnownErrors, testConstantAndZeroTargets, testPolynomialTrainPredict, testFourCandidateAPIs, testMLCellRejectsHolesAndExtrapolation, testMissingModelAndGate.

PHASE 13 - ML surrogate training and evaluation

Status: NOT EXECUTED

Files: matlab/ml/ml_train_surrogates.m, matlab/ml/ml_fit_candidate.m, matlab/ml/ml_apply_candidate.m, matlab/ml/ml_predict.m, matlab/ml/ml_metrics.m, matlab/ml/ml_diagnostic_report.m, matlab/ml/ml_learning_curve.m.

Dependencies: Phase 12, Statistics and Machine Learning Toolbox.

Included test suites: TestML.m.

Actually executed / reviewed: Static audit checked gate and validation-selection-before-test ordering. No training or performance benchmark executed.

Known limitations and unresolved issues: No trained MAT files or accuracy values. ANN uses fitrnet per output.

Test cases in the applicable shared suites:

- TestML.m: testSplitDeterministicAndDisjoint, testDuplicateInputsRejected, testMetricsKnownErrors, testConstantAndZeroTargets, testPolynomialTrainPredict, testFourCandidateAPIs, testMLCellRejectsHolesAndExtrapolation, testMissingModelAndGate.

PHASE 14 - MATLAB GUI

Status: NOT EXECUTED

Files: gui/EngineAnalyser_v1.m, gui/gui_build.m, gui/gui_action.m, gui.

Dependencies: Phases 1-13, MATLAB desktop.

Included test suites: TestGUI.m.

Actually executed / reviewed: Host audit checks callback action registry and source references. No desktop walkthrough.

Known limitations and unresolved issues: Programmatic AppBase .m instead of .mlapp Design

View artifact.

Test cases in the applicable shared suites:

- TestGUI.m: testEightTabsAndKaveriWarning.

PHASE 15 - Full integration and reports

Status: WARNING

Files: run_project.m, verify_project.m, scripts/audit_dependencies.m, scripts/run_phase_tests.m, simulink, matlab/reporting, docs, reports.

Dependencies: All previous phases, Simulink for model execution.

Included test suites: TestIntegration.m, TestSimulink.m, TestGUI.m.

Actually executed / reviewed: 28 host checks passed. Documentation/PDF and archive checks are recorded separately. Full MATLAB/Simulink integration NOT EXECUTED.

Known limitations and unresolved issues: Text .mdl plus .slx builder; no generated .slx, real benchmark or fabricated publication history.

Test cases in the applicable shared suites:

- TestIntegration.m: testPersistenceAndLog, testFailedPointLogsWithoutPerformance, testLogImmutable, testArchitectureGridCounts, testGridValidityMaskAndTemperatureFields, testCompatibleComparison, testIncompatibleComparisonRejected, testAllMissionPresets, testUnverifiedReferencesCannotPass, testReportingContainsProvenance, testOutputUnitConversion, testHTMLTextEscaped.
- TestSimulink.m: testBuiltModelsMatchBackend.
- TestGUI.m: testEightTabsAndKaveriWarning.